

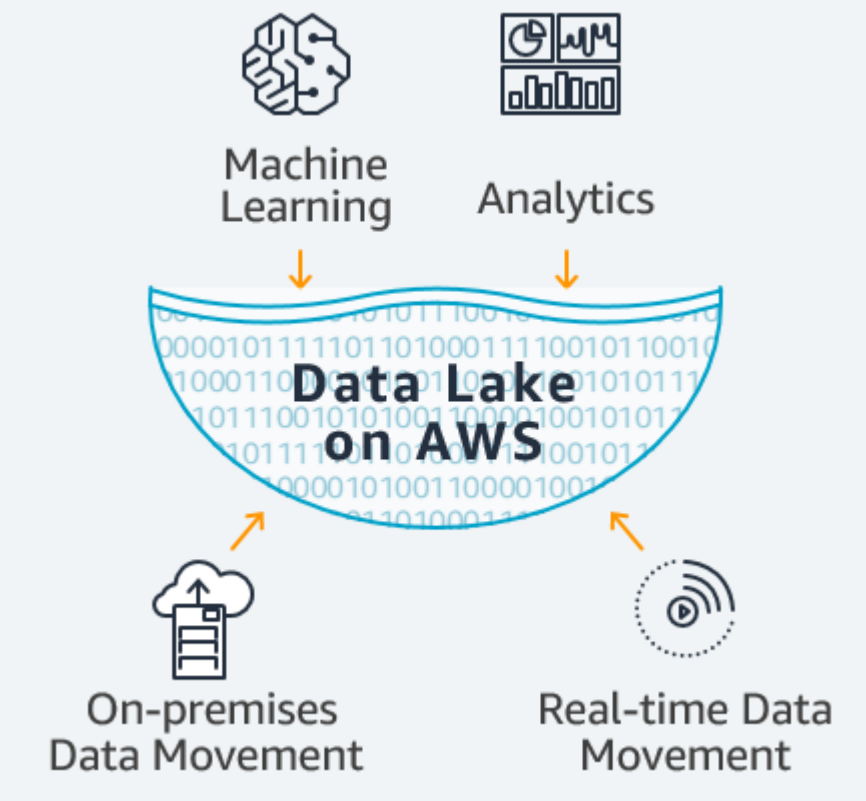


Proveedores de soluciones: AWS

Rafael Pastor Vargas
Andrés Duque Fernández
SCC-ETSI Informática-UNED

https://aws.amazon.com/es/big-data/datalakes-and-analytics/

BIG DATA EN AWS: DATA LAKES Y ANALÍTICA

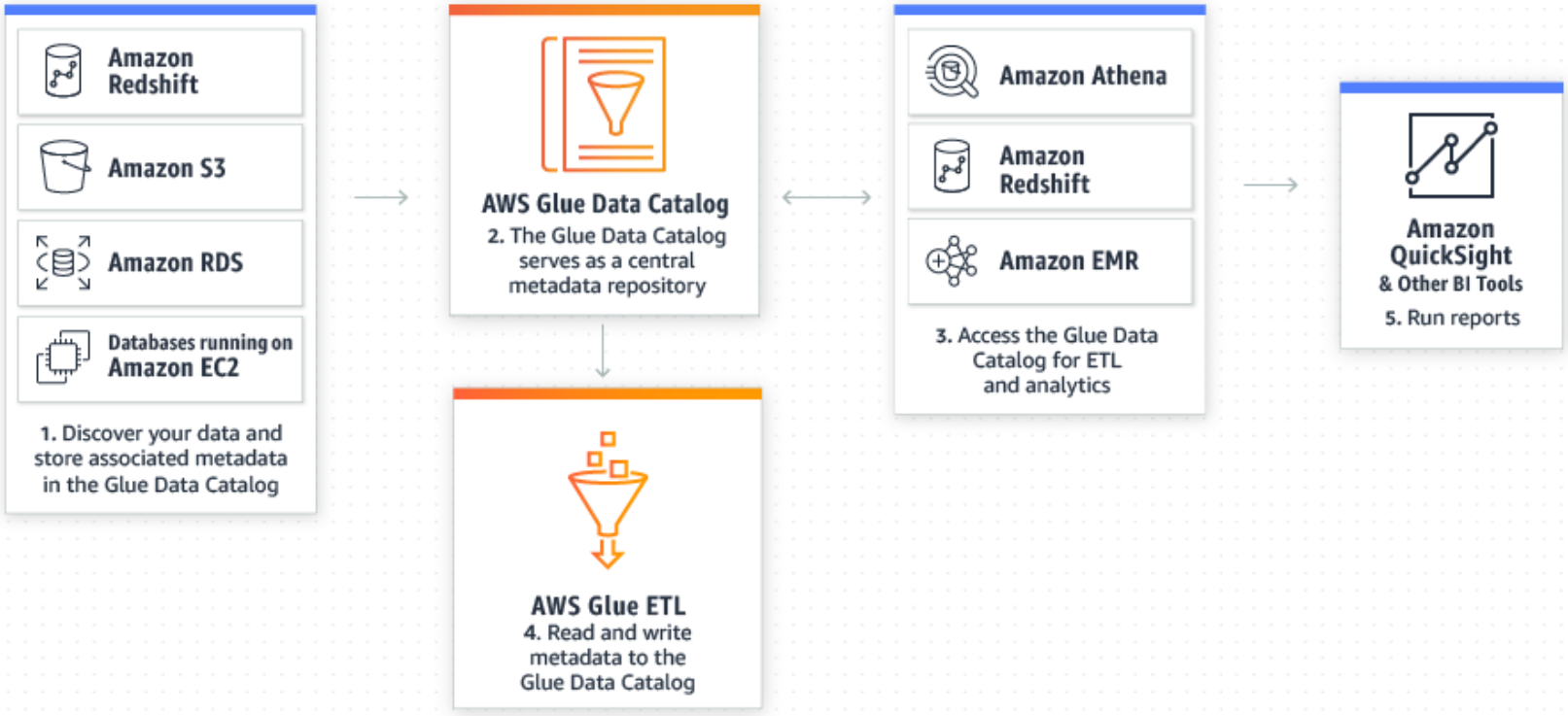


https://aws.amazon.com/es/blogs/big-data/

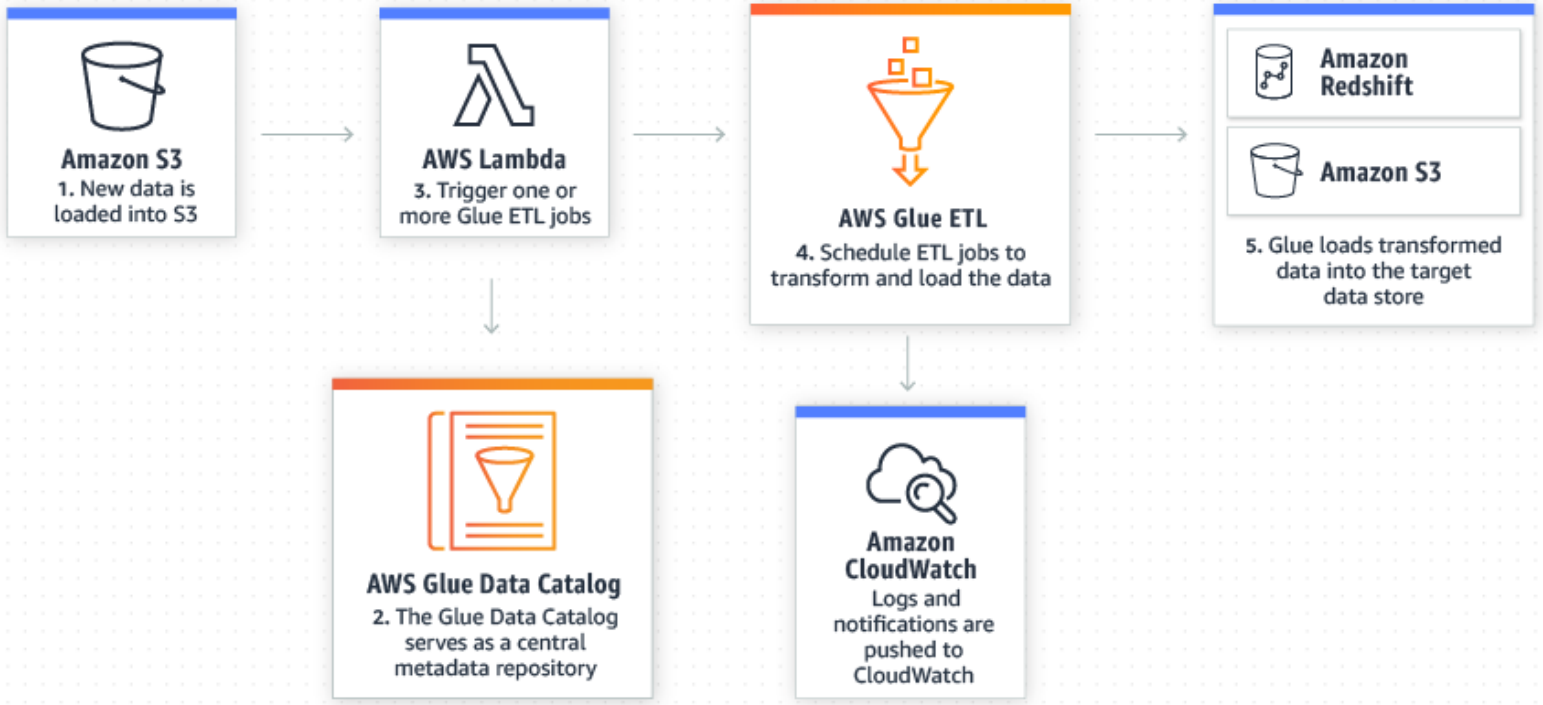
CASOS DE USO (I): CONSULTAS EN UN LAGO DE DATOS DE AMAZON S3



CASOS DE USO (II): ANÁLISIS DE DATOS DE REGISTROS EN SU ALMACÉN DE DATOS



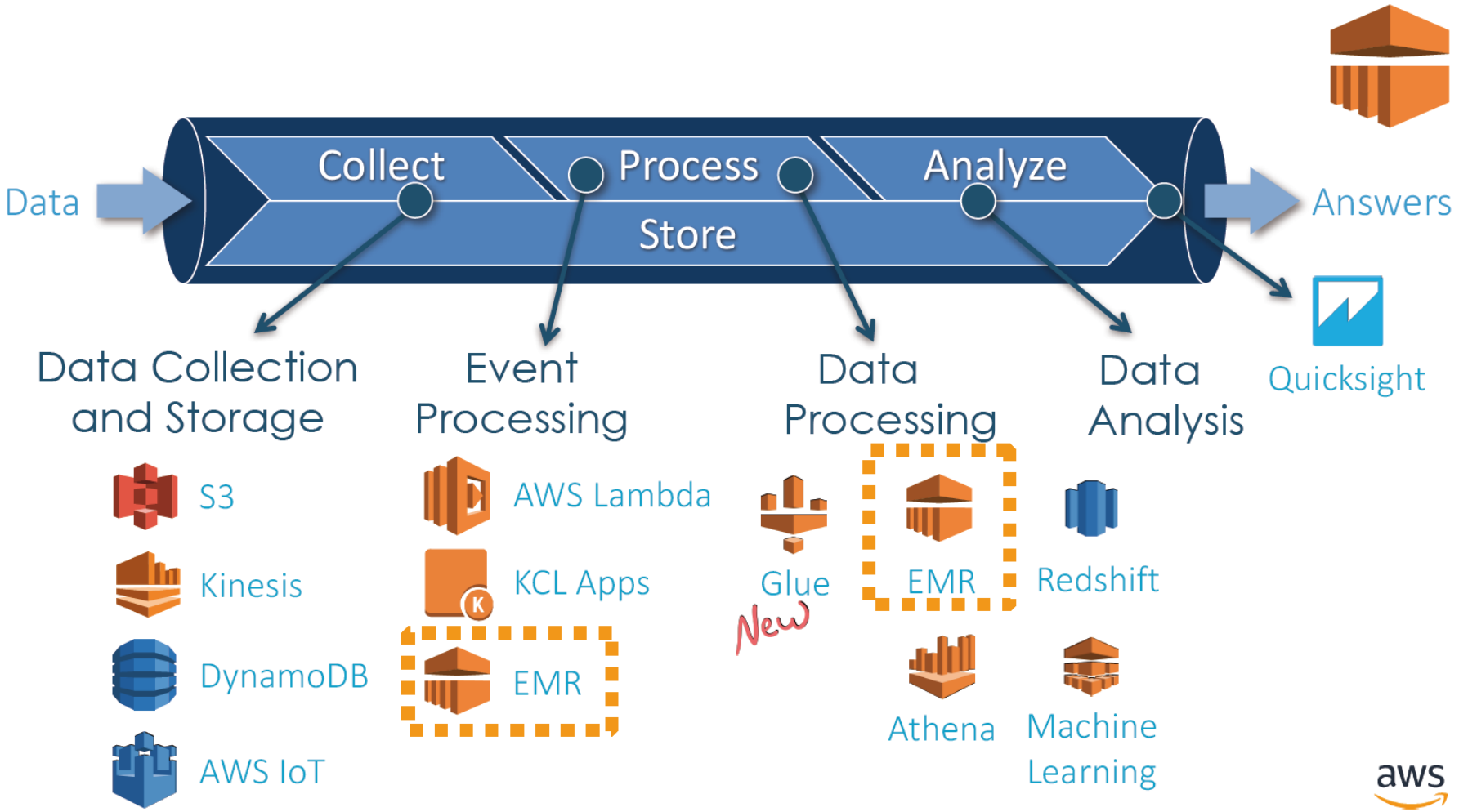
CASOS DE USO (III): CANALIZACIONES DE ETL DETERMINADAS POR EVENTOS



https://aws.amazon.com/es/big-data/

AMAZON WEB SERVICES (AWS)

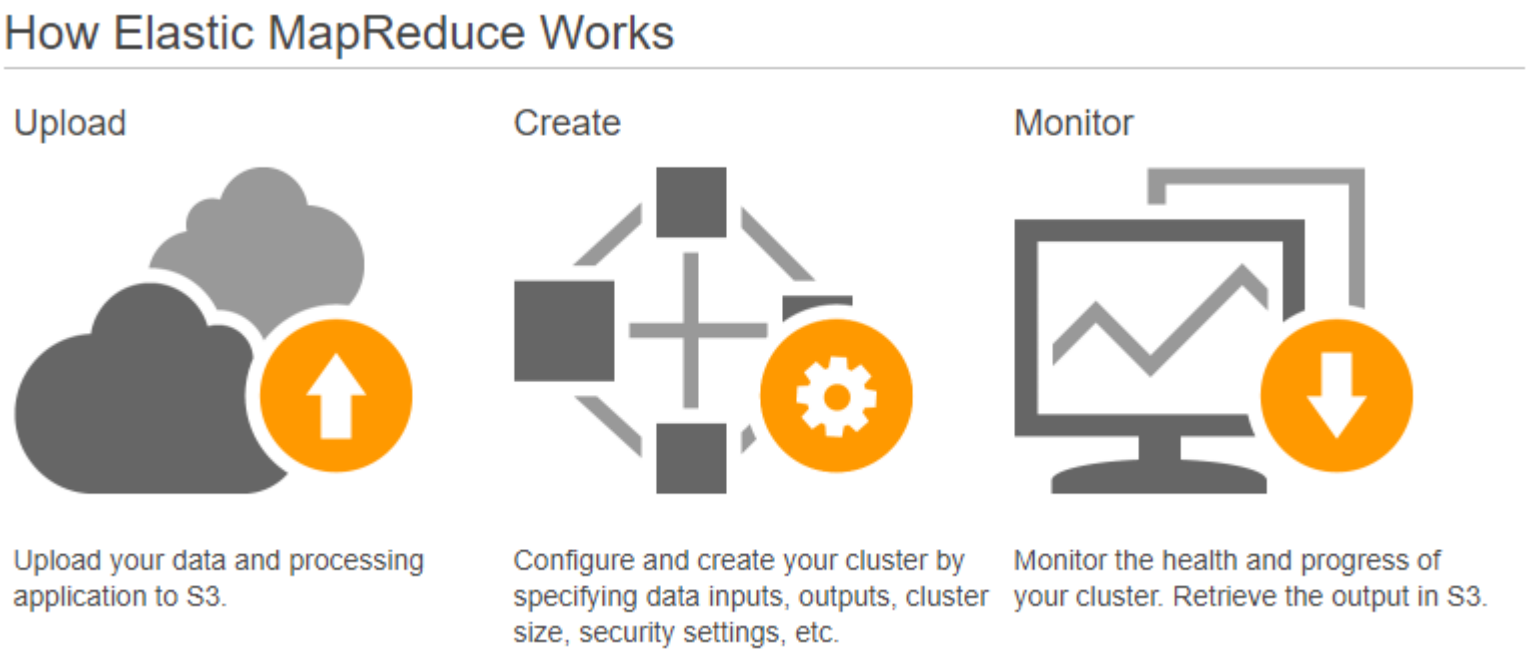
Big data en AWS



aws

https://aws.amazon.com/es/emr/

AMAZON EMR



https://aws.amazon.com/es/documentation/emr/

EMR: INFRAESTRUCTURA



Applications
Hive, Pig, Spark SQL/Streaming/ML, Mahout, Sqoop

Batch *MapReduce*	**Interactive** *Tez*	**In Memory** *Spark, Flink*

YARN
Cluster Resource Management

Storage
S3 (EMRFS), HDFS

HBase / Phoenix

Presto



Hue *(SQL Interface/Metastore Management)*
Zeppelin *(Interactive Notebook)*
Ganglia *(Monitoring)*
HiveServer2/Spark Thriftserver *(JDBC/ODBC)*



Amazon EMR service

	5.20.0 Dec. 2018	5.21.0 Feb. 2019	5.22.0 March 2019	5.23.0 April 2019	5.24.0 June 2019
Hadoop	2.8.5	2.8.5	2.8.5	2.8.5	2.8.5
Flink	1.6.2	1.7.0	1.7.1	1.7.1	1.8.0
Ganglia	3.7.2	3.7.2	3.7.2	3.7.2	3.7.2
HBase	1.4.8	1.4.8	1.4.9	1.4.9	1.4.9
Hive & HCatalog	2.3.4	2.3.4	2.3.4	2.3.4	2.3.4
Hue	4.3.0	4.3.0	4.3.0	4.3.0	4.4.0
JupyterHub	0.9.4	0.9.4	0.9.4	0.9.4	0.9.6
Livy	0.5.0	0.5.0	0.5.0	0.5.0	0.6.0
Mahout	0.13.0	0.13.0	0.13.0	0.13.0	0.13.0
MXNet	1.3.1	1.3.1	1.3.1	1.3.1	1.4.0
Oozie	5.0.0	5.0.0	5.1.0	5.1.0	5.1.0
Phoenix	4.14.0	4.14.0	4.14.1	4.14.1	4.14.1
Pig	0.17.0	0.17.0	0.17.0	0.17.0	0.17.0
Presto	0.214	0.215	0.215	0.215	0.219
Spark	2.4.0	2.4.0	2.4.0	2.4.0	2.4.2
Sqoop	1.4.7	1.4.7	1.4.7	1.4.7	1.4.7
Tensorflow	1.12.0	1.12.0	1.12.0	1.12.0	1.12.0
Tez	0.9.1	0.9.1	0.9.1	0.9.1	0.9.1
Zeppelin	0.8.0	0.8.0	0.8.1	0.8.1	0.8.1
Zookeeper	3.4.13	3.4.13	3.4.13	3.4.13	3.4.13
AWS SDK for Java	1.11.461	1.11.479	1.11.510	1.11.519	1.11.546

SPOT INSTANCES

- https://aws.amazon.com/es/ec2/spot/use-case/emr/

EJEMPLOS Y LABS

- https://aws.amazon.com/es/training/self-paced-labs/
  - Qwiklabs (https://amazon.qwiklabs.com/)
    - Analyze Big Data with Hadoop
- Tutoriales
  - https://aws.amazon.com/es/emr/getting-started/?nc=sn&loc=5
- Ejemplos avanzados
  - https://github.com/awslabs/s3-inventory-usage-examples
  - https://github.com/awslabs/deeplearning-emr

EJERCICIO 1

- https://aws.amazon.com/es/getting-started/projects/analyze-big-data/



*Don't forget to terminate your EMR cluster and Amazon S3 bucket after you complete the project to avoid additional charges

AMAZON KINESIS

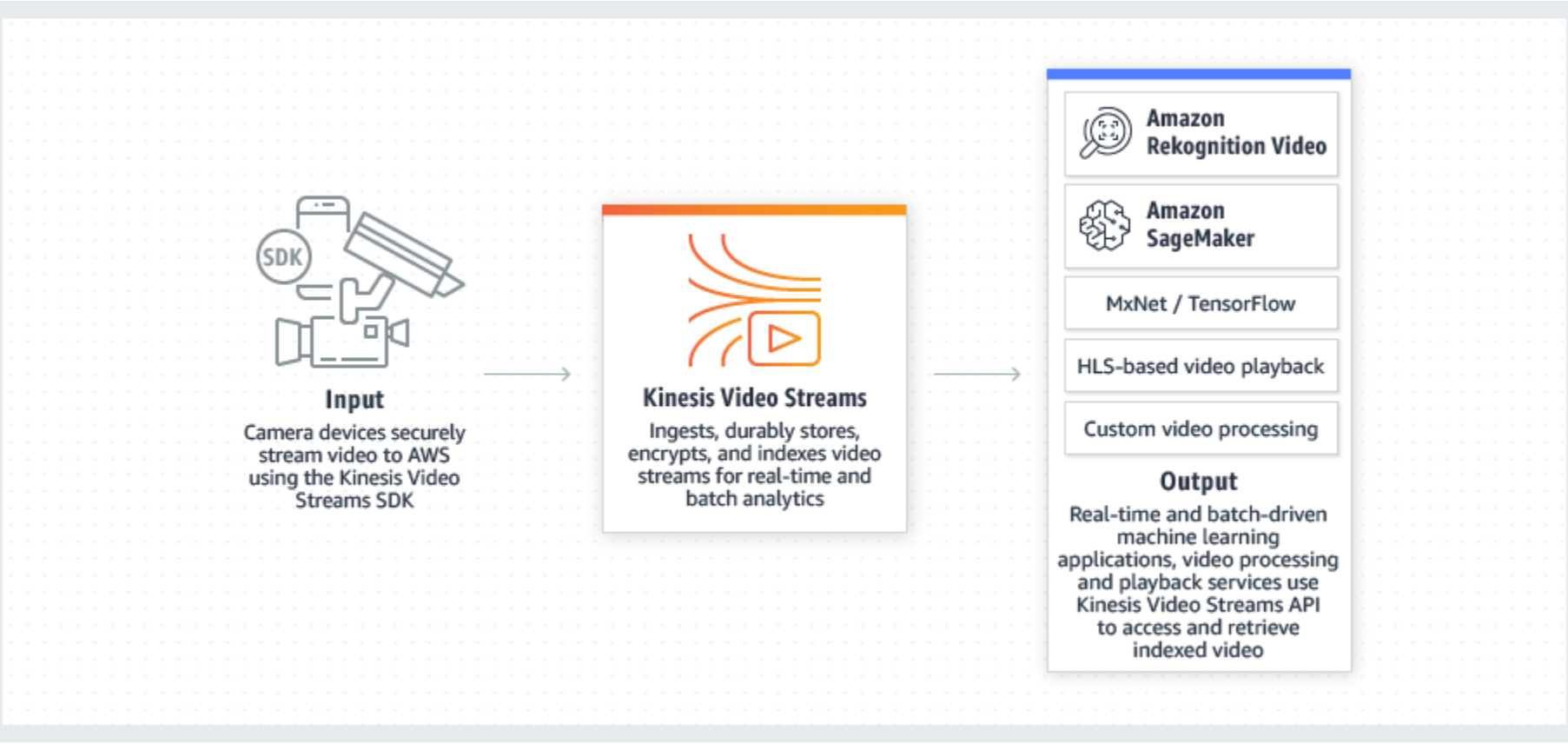
- Procesamiento Real-time
- Alta Eficiencia
- Escalable y Gestionado
- Integración con EMR, S3, Redshift, DynamoDB, etc.



Amazon Kinesis

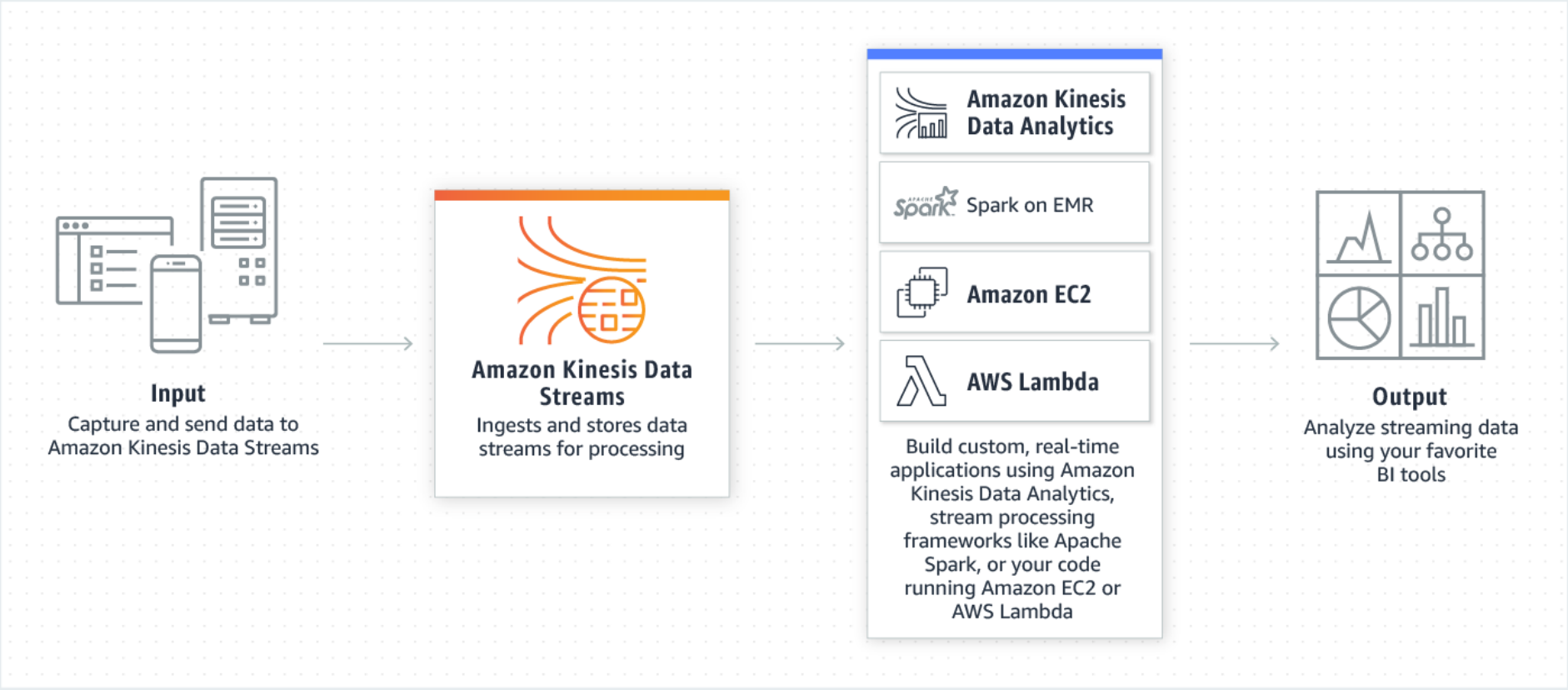
TIPOS (I)

- Amazon Kinesis Video Streams



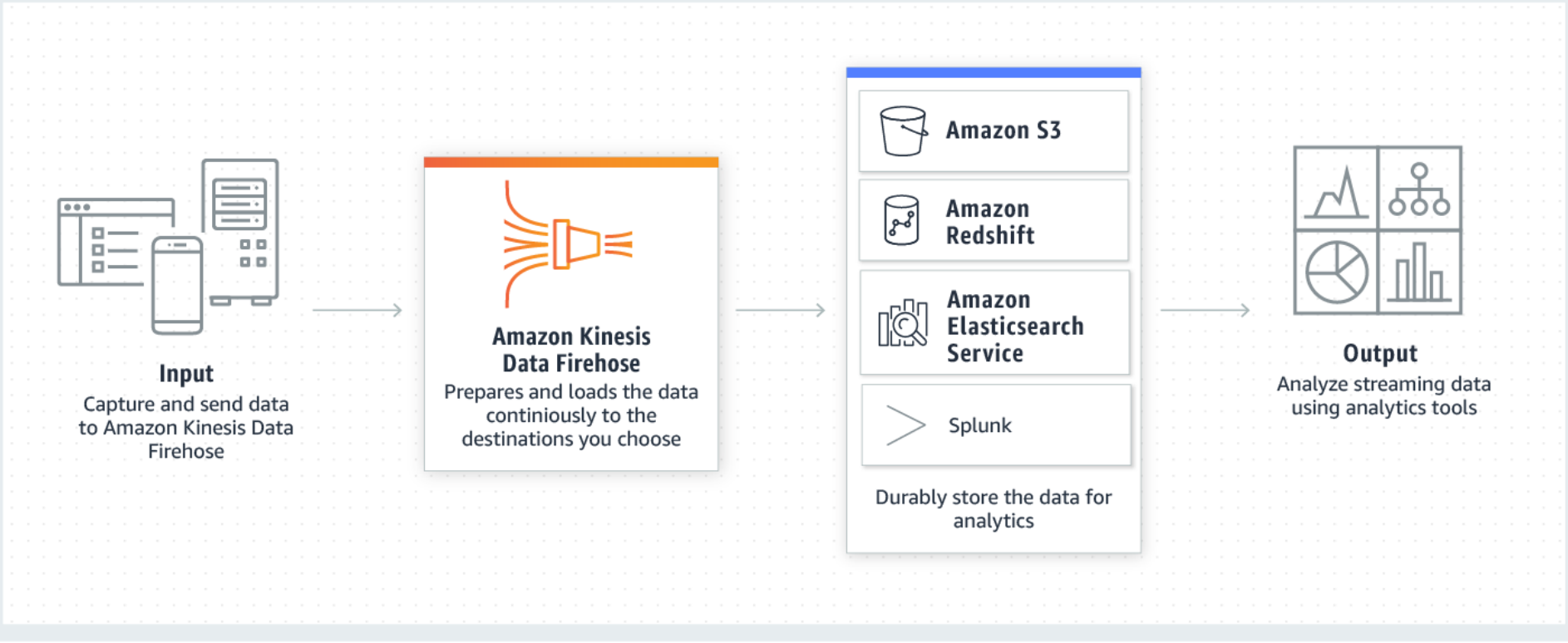
TIPOS (II)

- Amazon Kinesis Data Streams



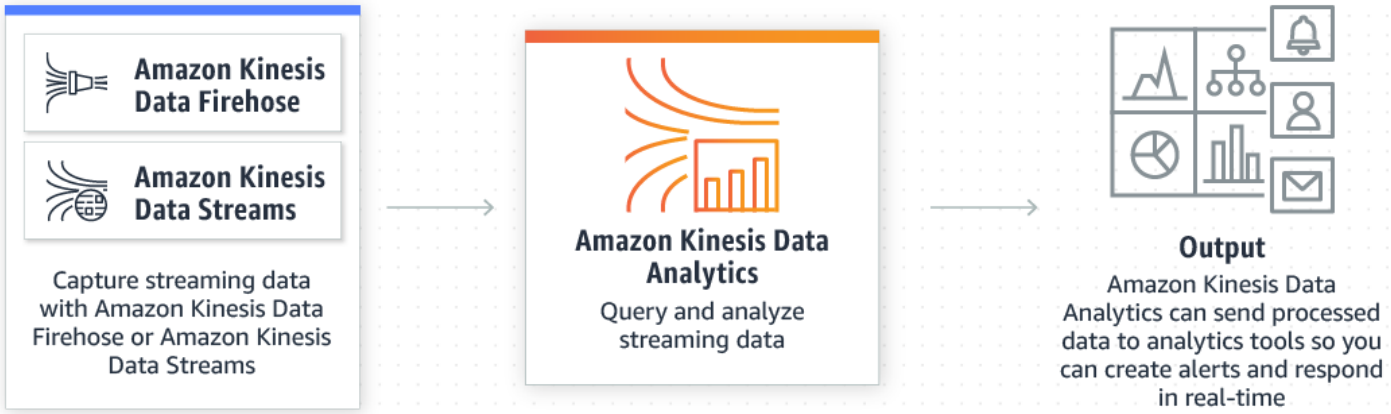
TIPOS (III)

- Amazon Kinesis Data Firehose



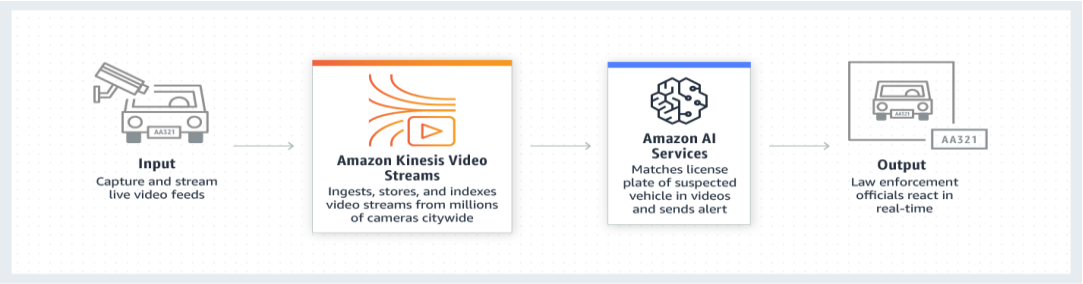
TIPOS (IV)

- Amazon Kinesis Data Analytics



- https://aws.amazon.com/es/kinesis/

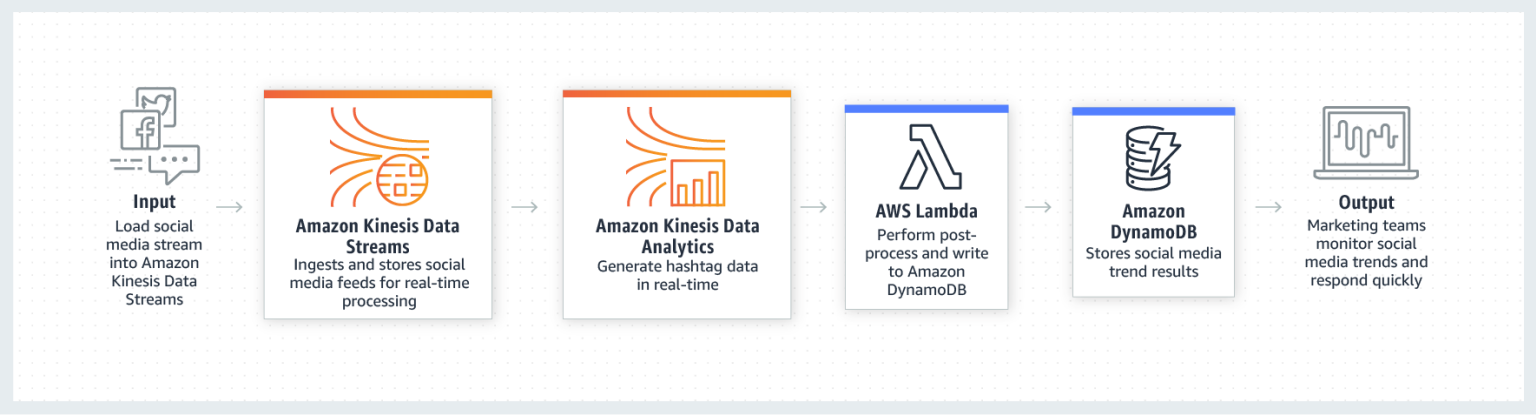
CASOS DE USO



VERITONE.



Zillow



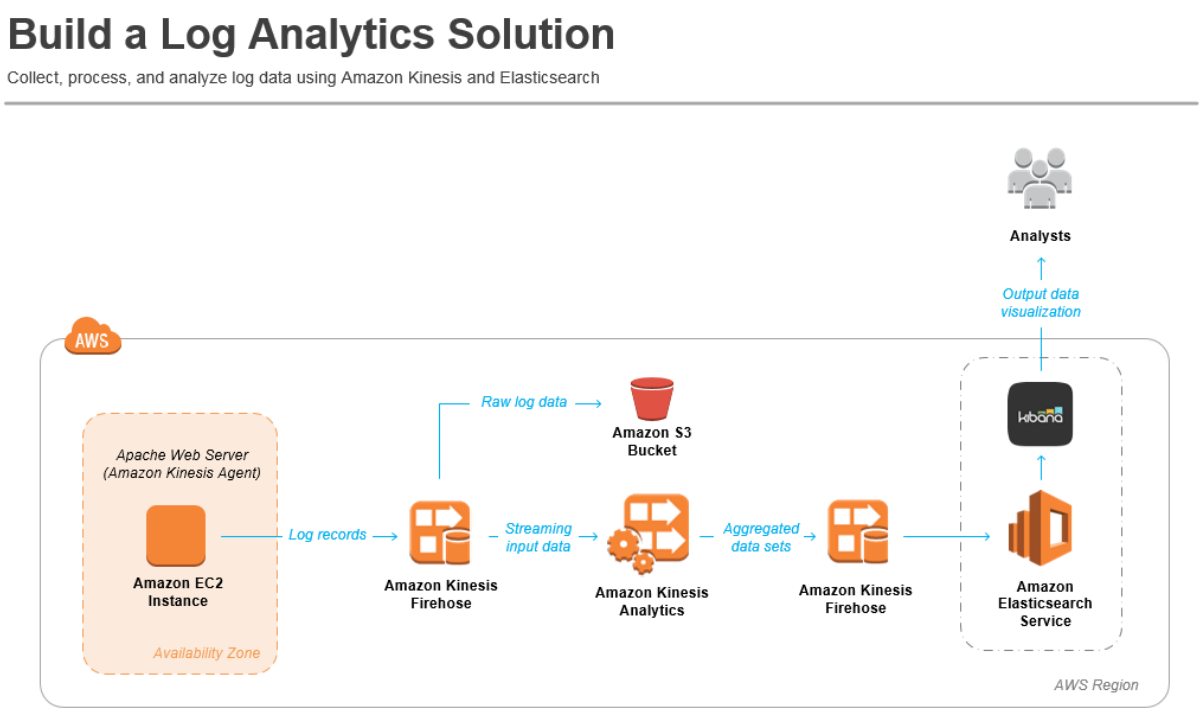
NETFLIX

EJERCICIOS Y LABS

- https://aws.amazon.com/es/training/self-paced-labs/
  - Qwiklabs (https://amazon.qwiklabs.com/)
    - Introduction to Amazon Kinesis Firehose
- Tutoriales
  - https://aws.amazon.com/es/kinesis/data-streams/getting-started/
- Ejemplos avanzados
  - https://github.com/awslabs/real-time-analytics-spark-streaming
  - https://github.com/awslabs/real-time-iot-device-monitoring-with-kinesis
  - https://github.com/awslabs/real-time-web-analytics-with-kinesis

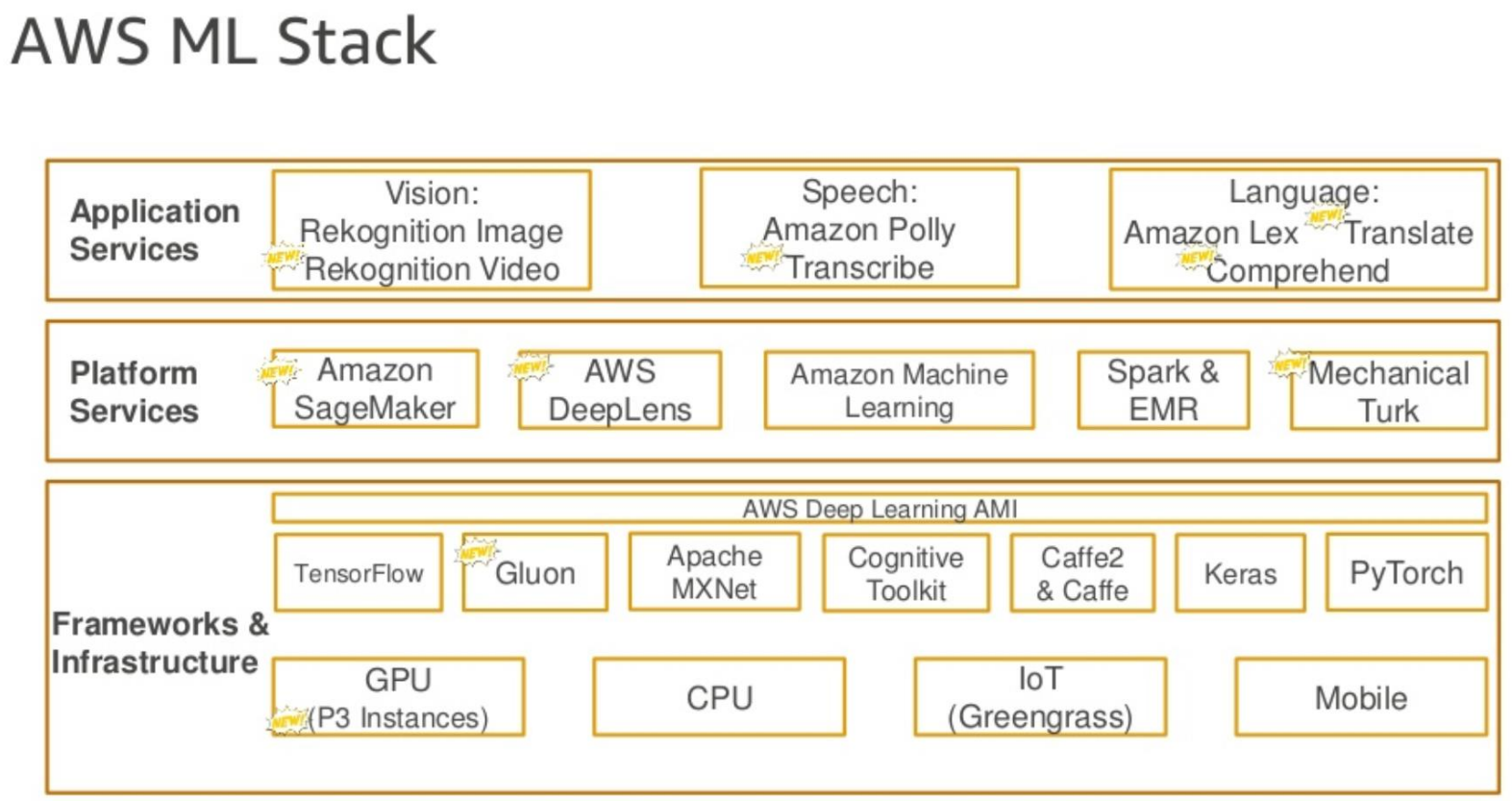
EJERCICIO 2

- https://aws.amazon.com/es/getting-started/projects/build-log-analytics-solution/



amazon web services

AWS MACHINE LEARNING



- Fuente: https://es.slideshare.net/AmazonWebServices/deep-learning-using-caffe2-on-aws-mcl313-reinvent-2017

AWS ML : INFRAESTRUCTURA

- Basada en EC2 (Containers y VM) → AMI

TensorFlow · mxnet · GLUON · Chainer · Caffe

theano · PYTORCH · K · The Microsoft Cognitive Toolkit · Caffe2

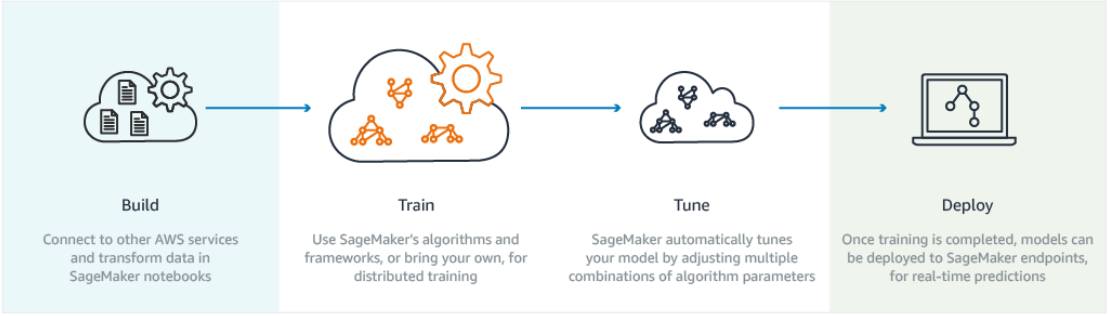


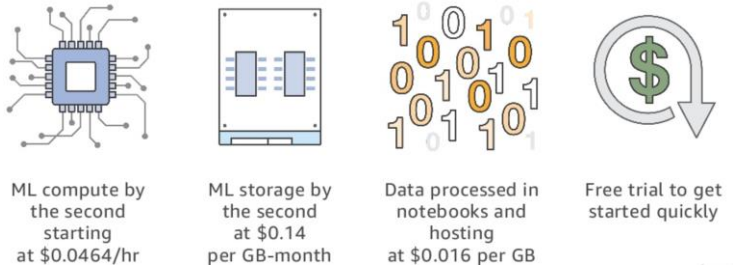
- Up to eight NVIDIA Tesla V100 GPUs
- 1 PetaFLOP of computational performance – *14x better than P2*
- 300 GB/s GPU-to-GPU communication (NVLink) – *9X better than P2*
- 16GB GPU memory with 900 GB/sec peak GPU memory bandwidth

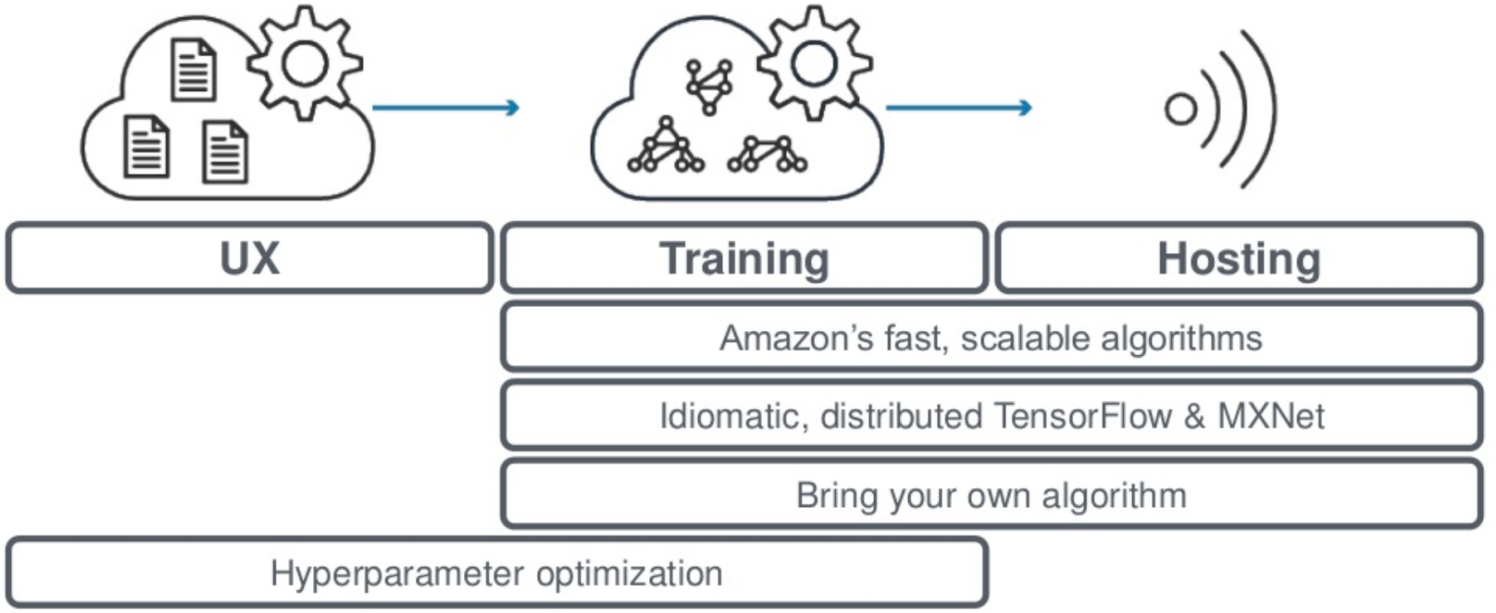
Instancias con GPU NVIDIA **Informática eficiente** intel **FPGA bajo demanda**

- https://es.slideshare.net/AmazonWebServices/introducing-amazon-sagemaker-aws-online-tech-talks-88664807

AWS ML: SAGEMAKER







AWS ML : SERVICIOS

